



Short communication

Selective peroxidative conversion of ethylbenzene catalyzed by Mn(II) complex under solvent-free conditions

Zohreh Razmara^{*}, Fatemeh Razmara, Somaye Shahraki

Department of Chemistry, University of Zabol, P.O. Box 98613-35856, Zabol, Iran

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ABSTRACT

An inorganic complex of Mn(II) formulated as $[\text{Mn}(\text{pzo})_2(\text{H}_2\text{O})_2]$ where pzo = Pyrazonic acid has been synthesized using two different methods; under ultrasonic irradiation as precipitate (**MS**) and by reflux as single crystal (**MR**). The structure of **MR** was characterized by single crystal X-ray diffraction (SC-XRD). The results of SC-XRD analysis exhibited that **MR** crystallizes in space group Fdd2 of an orthorhombic system and the centers of Mn^{2+} adopted a distorted octahedral geometry. **MS** and **MR** complexes were characterized by elemental analysis (CHN), atomic absorption spectroscopy (AAS), inductively coupled plasma optical emission spectrometry (ICP-OES), energy-dispersive X-ray spectroscopy (EDXS or EDS), X-ray photoelectron spectroscopy (XPS), Fourier-transform infrared spectroscopy (FT-IR), powder X-ray diffraction (PXRD), thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), scanning electron microscopy (SEM), and Brunner-Emmet-Teller (BET). The results of CHN, AAS, ICP-OES, EDS, XPS, FT-IR and TGA/DTA analyses showed that **MR** and **MS** complexes have the same structures. The results of PXRD, SEM, and BET analysis showed that the **MR** and **MS** have different particle size, morphology, and textural properties. The **MR** and **MS** were used as homogeneous catalysts in the solvent-free oxidation of ethylbenzene (EB) to acetophenone (AP) and benzaldehyde (BA) in the presence of *tert*-butyl hydroperoxide (TBHP) as an oxidant. Solvent-free oxidation of ethylbenzene is an environmentally friendly method since it does not need a solvent. The maximum conversion efficiency in the optimum operating conditions for **MS** and **MR** was 86.5 and 71.5 %, respectively. Finally, a free radical mechanism was proposed for the oxidation of ethylbenzene in the presence of **MS** and **MR**.

1. Introduction

Coordination complexes play vital role in adsorption and selective separation of materials, biological system, catalytic reactions, water oxidation, oxidation of hydrocarbons, photoluminescent, and as high-performance pH probe [1–5]. The use of hydrocarbons derived from oil and natural gas has become an important and significant scientific challenge for managing the global economy [6,7]. Hydroxylation, epoxidation, organoelectrophotocatalytic generation, electrophilic spirocyclization, selective oxidation of hydrocarbons, etc. are known as the most important processes that enables the conversion of hydrocarbons into high-value chemicals [8–11]. Although alkanes are abundant enough to be considered for use as starting materials in large-scale chemical processes, they are so inert that special conditions are required to convert them into molecules with useful functional groups [12]. Activation of C—H bonds in alkanes is really difficult and two important challenges related to these reactions are: (1) low selectivity,

due to the large number of C—H bonds with similar reactivity and bond dissociation energies, (2) the chemical inertness of C—H bonds of hydrocarbons, due to their low polarity [13,14]. Catalysts play an effective role in the selective activation of the C—H bonds, and in controlling the reaction pathway to improve the selectivity of hydrocarbons. The basis of modern chemical industry technology is the catalytic processes developed to convert alkanes into high value-added chemicals. Many studies have been reported on the activation of the C—H bonds of alkanes by organometallic and/or inorganic complexes [15,16]. In these reactions, a transition metal complex or its fragment such as a ligand with redox capability enters into the reactions [17]. Some complexes contain redox-active ligands that have the ability to transfer electrons to the metal center, which gives them undeniable importance [18]. Transition metal compounds are very important due to their greater stability and structural integrity. Recently, catalytic reactions based on transition metal complexes such as Mn, Cu, Fe, Ru, Pt, Pd, etc. are effectively used. Many of these reactions involve processes in which transition metal

^{*} Corresponding author.

E-mail address: razmara@uoz.ac.ir (Z. Razmara).

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